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Foreword

This Technical Report has been produced by the 3rd Generation Partnership Project (3GPP).

The present document is part of a multi-part TR as described below:

- Part 1: LI for IMS based services. See TR 33.929-1 [10].
- Part 2: LI for IMS based STIR/SHAKEN. See TR 33.929-2 [11].
- Part 3: LI for messaging services. See TR 33.929-3 [12].
- Part 4: LI for data in 5G core. See TR 33.929-4 [13].
- Part 5: LI for ID association caching. See TR 33.929-5 [14].
- Part 6: LI for IMS based RCS. See TR 33.929-6 [15].
- Part 7: LI location acquisition capabilities (present document).
- Part 8: LI for MMS. See TR 33.929-8 [16].
- Part 9: LI for PTC. See TR 33.929-9 [17].

In Part 1, the illustrations of LI for IMS-based services focus on the LI aspects of IMS sessions which include the architecture topologies and the call flows covering basic sessions, redirected sessions, target non-local ID, conferencing, roaming (local break-out and home-routed roaming).

In Part 2, the illustrations of LI for IMS based STIR/SHAKEN focus on various STIR/SHAKEN related LI reporting scenarios.

In Part 3, the illustrations of LI for messaging services focus on SMS over NAS and SMS over IP.

In Part 4, the illustrations of LI for data in 5G core focus on the LI aspects of data interception focussing on the PDU session related events including the handover scenarios.

In Part 5, the illustrations of LI for ID association caching focus on the capabilities used to provide the temporary identity to permanent identity (and vice-versa) associations known to the CSP network. The illustrations include the architectural concepts and few examples of use-cases encountering various timing scenarios.

In Part 6, the illustrations of LI for IMS based RCS focus on the architecture topologies and the call flows when the RCS service is offered by the CSP or by a Third Party Provider.

In Part 7 (present document), the illustrations of LI location acquisition capabilities focus on the conceptual overview and the flow diagrams for location reporting, LALS and location acquisition.

In Part 8, the illustrations of LI for MMS focus on the conceptual overview and the flow diagrams for MMS.

In Part 9, the illustrations of LI for PTC focus on the conceptual overview and the flow diagrams for MCPTT and PoC, the two commonly referred to as Push to Talk over Cellular (PTC).

Introduction

Unlike the previous generation of LI technical specifications, the latest LI technical specifications (TS 33.126 [2], TS 33.127 [3], TS 33.128 [4]) contain only the normative part of the requirements. However, the implementers may need additional information such as architectural diagrams, conceptual scenarios, flow-diagrams and additional background information to better understand the requirements in those specifications. The present document collects the relevant informative annexes from the previous generation of LI technical specifications and then expands them to include the similar level of details as for latest technologies such as 5G.

1 Scope

The present document provides architectural diagrams, conceptual scenarios, flow-diagrams, examples, and other background information which can be useful to implement the LI functions defined in TS 33.126 [2], TS 33.127 [3] and TS 33.128 [4].

The present document covers the LI for Location Services that include the illustrations covering the conceptual overview and the flow diagrams for location reporting, LALS and Location Acquisition.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 33.126: "Lawful Interception requirements".

[3] 3GPP TS 33.127: "Lawful Interception (LI) architecture and functions".

[4] 3GPP TS 33.128: "Protocol and procedures for Lawful Interception (LI)".

[5] 3GPP TS 24.147: "Conferencing using the IP Multimedia (IM) Core Network (CN) subsystem; Stage 3".

[6] 3GPP TS 23.501: "System architecture for the 5G System".

[7] 3GPP TS 24.174: "Support of multi-device and multi-identity in the IP Multimedia Subsystem (IMS); Stage 3".

[8] 3GPP TS 23.228: "IP Multimedia Subsystem (IMS), Stage 2".

[9] 3GPP TR 33.928: "ADMF logic for provisioning Lawful Interception (LI)".

[10] 3GPP TR 33.929-1: "Lawful Interception (LI) implementation guidance; LI for IMS-based services".

[11] 3GPP TR 33.929-2: "Lawful Interception (LI) implementation guidance; LI for IMS based STIR/SHAKEN".

[12] 3GPP TR 33.929-3: "Lawful Interception (LI) implementation guidance; LI for messaging services".

[13] 3GPP TR 33.929-4: "Lawful Interception (Li) implementation guidance; LI for data in 5G core".

[14] 3GPP TR 33.929-5: "Lawful Interception (LI) implementation guidance; LI for ID association caching ".

[15] 3GPP TR 33.929-6: "Lawful Interception (LI) implementation guidance; LI for IMS based RCS".

[16] 3GPP TR 33.929-8: "Lawful Interception (LI) implementation guidance; LI for MMS".

[17] 3GPP TR 33.929-9: "Lawful Interception (LI) implementation guidance; LI for PTC".

3 Definitions of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1].

3.2 Symbols

For the purposes of the present document, the following symbols apply:

None.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

3GPP	3rd Generation Partnership Program
ADMF	Administration Function
CC	Content of Communication
CSCF	Call Session Control Function
CSP	Communication Service Provider
ID	Identity
LAF	Location Acquisition Function
LALS	Lawful Access Location Services
LARF	Location Acquisition Requesting Function
LEA	Law Enforcement Agency
LEMF	Law Enforcement Monitoring Facility
LI	Lawful Interception
LICF	Lawful Interception Control Function
LI_HI1	LI_Handover Interface 1
LI_HI2	LI_Handover Interface 2
LI_HILA	Lawful Interception Handover Interface Location Acquisition
LIPF	LI Provisioning Function
LI_X1	Lawful Interception Internal Interface 1
LI_X2	Lawful Interception Internal Interface 2
LI_X2_LA	Lawful Interception Internal Interface 2 Location Acquisition
LI_XLA	Lawful Interception Internal Interface Location Acquisition
LTE	Long Term Evolution
LTF	Location Triggering FunctionMCPTT Mission Critical Push To Talk
MDF	Mediation and Delivery Function
MDF2	Mediation and Delivery Function 2
MMS	Multimedia Message Service
NR	New Radio
P-CSCF	Proxy Call Session Control Function
PEI	Permanent Equipment Identifier
PoC	Push to talk over Cellular
POI	Point Of Interception
PTC	Push to Talk over Cellular
RCS	Rich Communication Suite
S-CSCF	Serving Call Session Control Function
SGW	Serving Gateway
SHAKEN	Signature-based Handling of Asserted information using toKENs
SMF	Session Management Function
STIR	Secure Telephony Identity Revisited
SUPI	Subscriber Permanent Identifier

TF	Triggering Function
UDM	Unified Data Management
xIRI	LI_X2 Intercept Related Information

4 Illustration of LI for location services

4.1 Overview

The LI for location services illustrated in the subsequent clauses include the following three parts:

- Location only LI reporting.
- Lawful access location services (LALS).
- Location acquisition.

There are also other cases already illustrated in TS 33.127 [3] and TS 33.128 [4].

For example, with the case referred to as embedded location reporting as defined in TS 33.127 [3], the IRIs delivered to the LEMF as a part of interception performed for other services may include the location if the warrant authorized such location reporting. In this case, the reported location is the location of the target UE.

The location reporting does not apply when the interception is performed for a target non-local ID. When the warrant prohibits the reporting of the location, any information that conveys the location is redacted from the IRI/CC delivered to the LEMF.

In all cases, the scope of the reported location may vary as defined in TS 33.127 [3] and TS 33.128 [4].

4.2 Location only LI reporting

4.2.1 General

Location only LI reporting refers to a reporting for a warrant authorizing only the target's location information.

Location only LI reporting does not apply to the target non-local ID.

The details on the scope of reported location, name of the related xIRIs/IRIs and the parameters used within those xIRI/IRI are defined in TS 33.127 [3] and TS 33.128 [4].

The present document uses the term Location Only xIRI and Location Only IRI to refer collectively to all of the xIRIs/IRIs used for the interceptions that deal with the location only LI reporting. For example, SeparatedLocationReporting is a Location Only xIRI/IRI.

The present document refers to the warrants issued for interceptions other than location only LI reporting as regular warrants and the associated interceptions as regular interceptions.

4.2.2 Overview of the concept

Figure 4.2.2-1 provides an overview of the concepts of location only LI reporting, depicting two implementation options. In both options, the process begins when an LEA issues a warrant to the CSP for location only LI reporting on a target UE.

All of the LI functions and LI interfaces depicted in figure 4.2.2-1 are defined in TS 33.127 [3] and TS 33.128 [4]. The NF shown in figure 4.2.2-1 can be any NF within the CSP network that has an IRI-POI with access to the target's location information. Few examples of such NFs: AMF, MME, HSS, UDM, S-CSCF, P-CSCF.

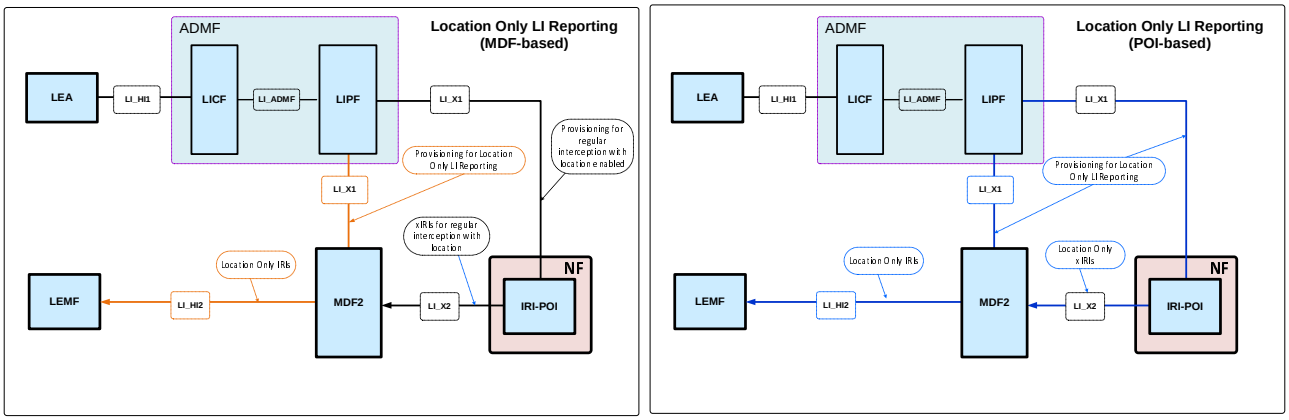


Figure 4.2.2-1: An overview of location only LI reporting

In the MDF-based implementation option, the IRI-POI is not provisioned for location only LI reporting (instead, the IRI-POI is provisioned for regular interception with location enabled). MDF2 is provisioned with location only LI reporting, so whenever the MDF2 receives an xIRI that has the location information, it constructs the Location Only IRI messages and delivers them to the LEMF.

As an example, if the MDF2 receives the IMSMessage xIRI record that includes the location information of the target, then the MDF2 would construct a SeparatedLocationReporting IRI message from the IMSMessage xIRI record and delivers the SeparatedLocationReporting IRI message to the LEMF.

In the POI-based implementation option, both the IRI-POI and the MDF2 are provisioned for location only LI reporting. In this option, the IRI-POI delivers the Location Only xIRIs to the MDF2 and the MDF2 would construct Location Only IRI messages to the LEMF.

As an example, whenever the IRI-POI in S-CSCF observes an event that would normally result in the generation of an IRIMessage xIRI record, if the IRIMessage xIRI record would have included the location information of the target, the IRI-POI would, instead, construct and deliver a SeparatedLocationReporting xIRI to the MDF2 and the MDF2 would construct the SeparatedLocationReporting IRI message and delivers it to the LEMF.

The AMFLocationUpdate xIRI/IRI which is used with the regular interception is also used as an xIRI/IRI with location only LI reporting. Therefore, with the MDF-based implementation option, when the MDF2 receives the AMFLocationUpdate xIRI record, it constructs the AMFLocationUpdate IRI message and delivers it to the LEMF. However, if another xIRI is received from the IRI-POI present in the AMF that has the location, then the MDF2 would construct a SeparatedLocationReporting IRI message from it and deliver the same to the LEMF. The similar concepts apply for the xIRIs received from the IRI-POIs present in the MME, UDM and HSS.

4.2.3 Location only LI reporting scenarios

4.2.3.1 General

As described in clause 4.2.2, there are two implementation options for location only LI reporting. This clause illustrates a few scenarios that govern those options.

4.2.3.2 One warrant

Figure 4.2.3.2-1 illustrates the use of POI-based implementation option for location only LI reporting. The LEA issues a warrant for authorizing only the target UE's location information (step P1).

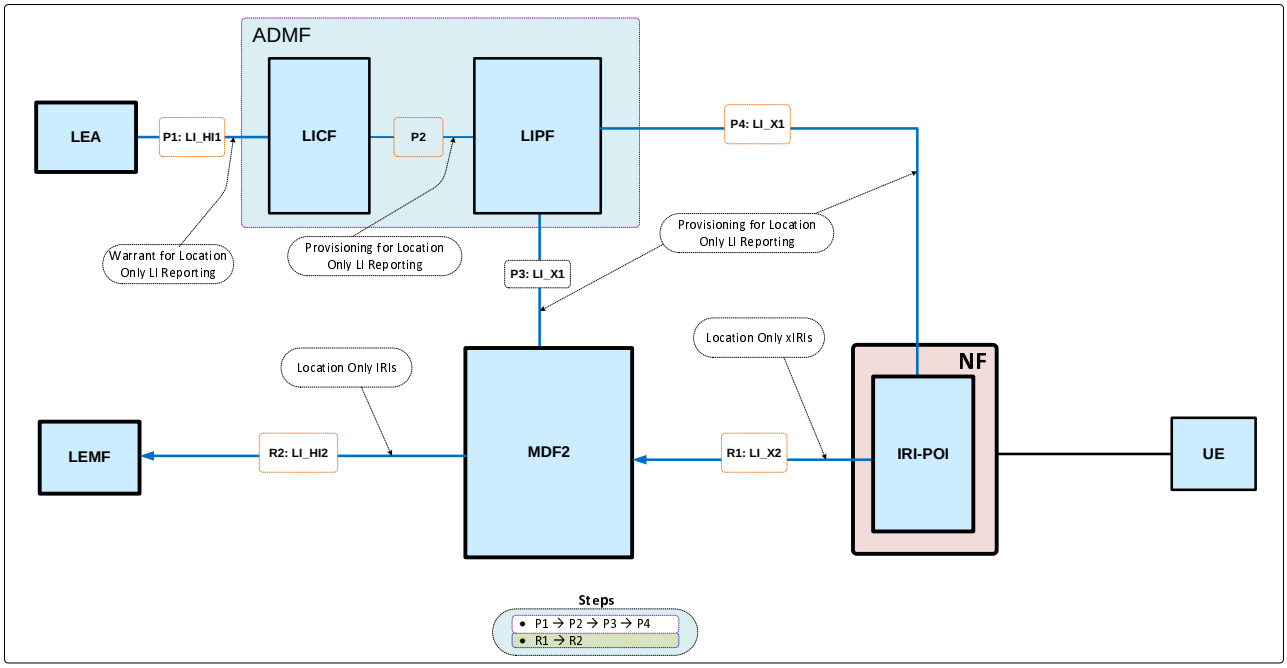


Figure 4.2.3.2-1: Location only LI reporting – one warrant (POI-based option)

The IRI-POI and the MDF2 are provisioned for location only LI reporting. In figure 4.2.3.2-1, the steps P2, P3 and P4 are the provisioning steps.

The IRI-POI generates the Location Only xIRIs and delivers them to the MDF2. The MDF2 constructs the Location Only IRIs and delivers them to the LEMF. In figure 4.2.3.2-1, the steps R1 and R2 show the Location Only xIRIs and IRIs respectively, generated for the warrant from LEA.

4.2.3.3 Two warrants, one target, scenario of MDF-based implementation option

Figure 4.2.3.3-1 illustrates the use of MDF-based implementation option for location only LI reporting.

LEA-1 issues a regular warrant for one of the services listed in TS 33.127 [3]/TS 33.128 [4] for the UE (step P1-1). The LEA-2 issues a location only LI reporting warrant on the same target UE (step P1-2).

The illustrated approach can be more suitable for the case where the provisioning for the regular warrant (from LEA-1) is done before the provisioning for the location only LI reporting warrant (from LEA-2). The provisioning for regular warrant is assumed to have the location reporting enabled.

However, the illustrated approach can still be used even for the case where the provisioning for the location only LI reporting (from LEA-2) is done first or when the provisioning of regular warrant (from LEA-1) if done first, does not enable the location reporting, by modifying the existing task at the POI for regular interception with location reporting enabled.

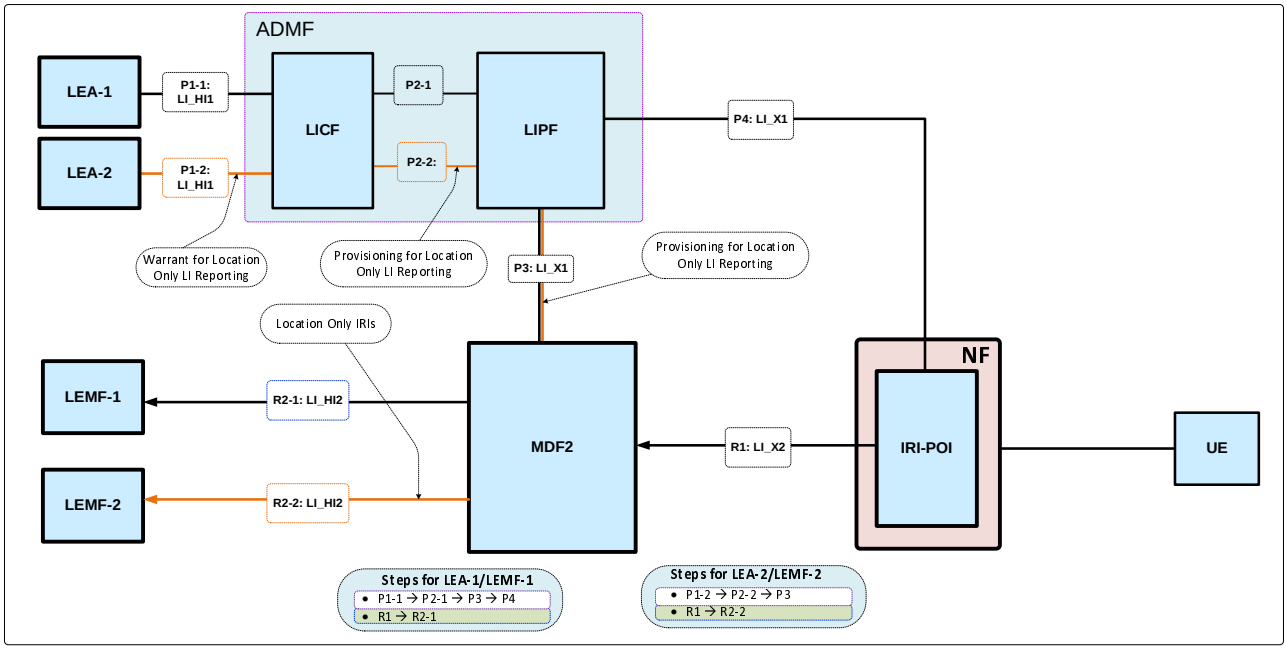


Figure 4.2.3.3-1: Location only LI reporting – two warrants, one target UE, scenario 1

As part of the handling of the regular warrant from LEA-1, the IRI-POI and the MDF2 are provisioned for regular interception. The MDF2 is provisioned with a new task with a new set of mediation details that include the LIID assigned LEA-1 and point to LEMF-1 as the delivery destination. In figure 4.2.3.3-1, the steps P2-1, P3 and P4 are the provisioning steps for the regular warrant from LEA-1.

The IRI-POI generates and delivers all xIRI records to the MDF2. The MDF2 constructs relevant IRI messages and delivers them to the LEMF-1. In figure 4.2.3.3-1, the steps R1 and R2-1 show the xIRIs and the IRIs respectively, generated for the warrant from LEA-1.

As part of the handling of the location only LI reporting warrant from LEA-2, the MDF2 is provisioned by updating the existing task with a new set of mediation details that include the LIID assigned to LEA-2, LocationOnly indication and point to the LEMF-2 as the delivery destination. The mediation details for LEA-1 are included as originally provisioned when the task was created. In figure 4.2.3.3-1, the P2-2, P3 are the provisioning steps for the location only LI reporting warrant from LEA-2. The provisioning of MDF2 is an update when a second warrant is received.

The IRI-POI continues to generate and deliver all xIRI records to the MDF2. The MDF2 constructs the relevant IRI messages and delivers them to the LEMF-1. When the xIRIs received from the IRI-POI includes the location information, the MDF2 constructs the Location Only IRI messages and delivers them to the LEMF-2. In figure 4.2.3.3-1, the steps R1 and R2-1 show the xIRIs and the IRIs respectively, generated for the warrant from LEA-1, and the steps R1 and R2-2 show the xIRIs and the IRIs respectively, generated for the warrant from LEA.

As illustrated, in this scenario, the two warrants are handled by a single task in the IRI-POI and as a single task with two sets of mediation details at the MDF2.

4.2.3.4 Two warrants, one target, scenario of POI-based implementation option

Figure 4.2.3.4-1 illustrates the use of POI-based implementation option for location only LI reporting.

LEA-1 issues a location only LI reporting warrant on the UE (step P1-1). The LEA-2 issues a regular warrant for one of the services listed in TS 33.127 [3]/TS 33.128 [4] for the target UE (step P1-2).

The illustrated approach can be more suitable for the case where the provisioning for the location only LI reporting warrant (from LEA-1) is done before the provisioning for the regular warrant (from LEA-2) or when the provisioning of regular warrant (from LEA-2) if done first, does not enable the location reporting.

However, the illustrated approach can still be used even for the case where the provisioning for the regular warrant (from LEA-2) with location enabled is done first even though such an implementation is not optimal.

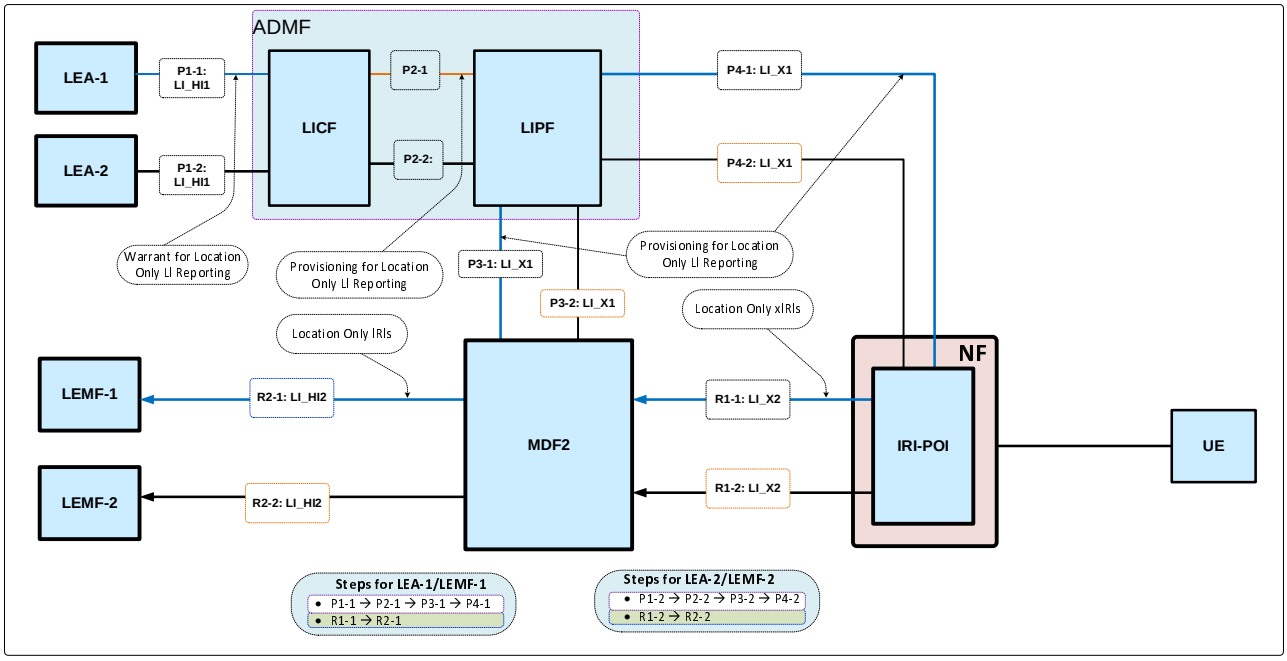


Figure 4.2.3.4-1: Location only LI reporting – two warrants, one target UE, scenario 2

As part of the handling of the location only LI reporting warrant from LEA-1, the IRI-POI and the MDF2 are provisioned for location only LI reporting. The MDF2 is provisioned with mediation details that include the LIID assigned to LEA-1, LocationOnly indication and point to LEMF-1 as the delivery destination. In figure 4.2.3.3-1, the steps P2-1, P3-1 and P4-1 are the provisioning steps for the location only LI reporting related warrant from LEA-1.

The IRI-POI generates and delivers the Location Only xIRI records to the MDF2 and the MDF2 constructs the Location Only IRI messages and delivers the same to LEMF-1. In figure 4.2.3.4-1, the steps R1-1 and R2-1 show the Location Only xIRIs and the IRIs respectively, generated for the location only LI reporting warrant from LEA-1.

As part of the handling of the regular warrant from LEA-2 on the same target UE, the IRI-POI and the MDF2 are provisioned with a new task for regular interception. In figure 4.2.3.4-1, the steps P2-2, P3-2 and P4-2 are the provisioning steps for the regular warrant from LEA-2.

The IRI-POI generates and delivers all xIRI records to the MDF2. The MDF2 constructs the relevant IRI messages and delivers them to the LEMF-2. In figure 4.2.3.4-1, the steps R1-2 and R2-2 show the xIRIs and the IRIs respectively, generated for the warrant from LEA-2.

As illustrated, in this scenario, the two warrants are handled as two independent interceptions even though the two warrants are on the same target.

4.3 LALS

4.3.1 General

The LALS provides LEA a capability to determine the physical location (also called position) of the target. There are two modes of LALS operation supported:

- On-demand location determination, called Target Positioning.
- Location determination synchronized with target interception events, called Triggered Location.

The Target Positioning is invoked by a designated HI1 warrant indicating the LALS service request. There are two options of Target Positioning supported: Immediate Positioning and Periodic Positioning. The former is intended to invoke a single positioning operation on the target, while the latter invokes a series of positioning operations with the periodicity specified in the warrant. There can be multiple intermediate location responses with the either of the two options.

With the Triggered Location, the target position is obtained and reported when certain events (e.g. registration, start/end of PDU session, start/end of IMS session) related to the target are detected and reported in the network in the course of execution of an interception warrant which includes the LALS triggered location option. While the reported interception events may carry the location information, the LALS triggering would enhance the reporting with the target position information, like geo-location, target velocity, etc.

LALS triggered location can be used with location only reporting as well. LALS does not apply to target non-local ID.

4.3.2 Overview of the architecture

In order to have the lawful access to the target's location, an LI specific LCS client referred to as LI-LCS Client introduced. The LI-LCS Client, a logical function, interacts with the LCS Server (e.g. GMLC) to obtain the target's location. Neither the present document nor the TS 33.127 [3]/33.128 [4] identify/define the NF that can house LI-LCS Client. The invocation of LALS mentioned in clause 4.3.1 is nothing but the invocation of location determination at the LI-LCS Client.

In order to support the invocation of LALS when certain events related to the target are detected, an LI specific logical function referred to as LALS Triggering Function (LTF) is introduced. The LTF would detect the specific events and invoke the LI-LCS Client for location determination. As defined in TS 33.127 [3], the LTF can be present either in the MDF2 or alongside an IRI-POI in a NF within the CSP domain.

Figure 4.3.2-1 illustrates an overview of the overall architecture for LALS depicting the modes of LALS invocation. All of the LI functions and LI interfaces depicted in figure 4.3.2-1 are defined in TS 33.127 [3] and TS 33.128 [4].

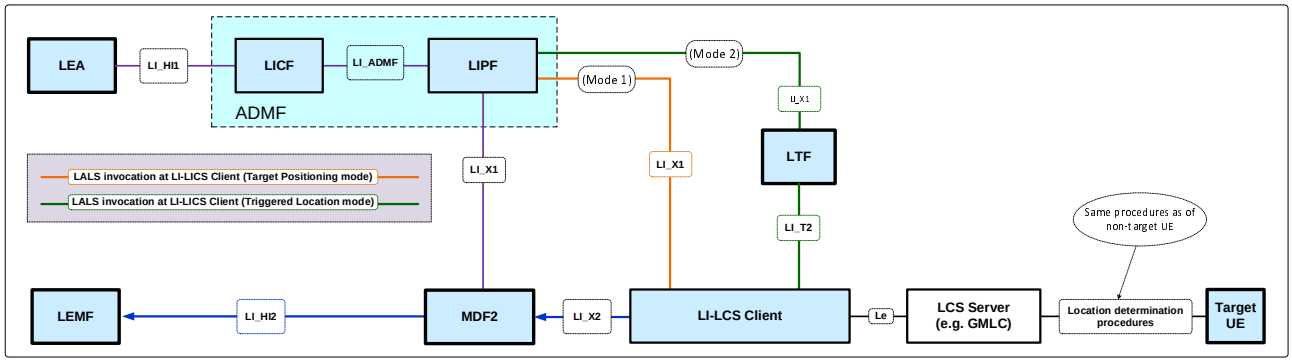


Figure 4.3.2-1: An overview of LALS

In the Target Positioning mode, the LALS is invoked in the LI-LCS Client by the LIPF over LI_X1. The LI-LCS Client interacts with the LCS Server (e.g. GMLC) to obtain the location either immediately or periodically as instructed by the LIPF. The LCS Server may return the target UE's intermediate location multiple times.

In the Triggered Location mode, the LIPF provisions the LTF for LALS over LI_X1. The LTF then invokes the LALS in the LI-LCS Client over LI_T2 when certain events related to the target UE are detected.

With both modes, in support of LALS, the LI-LCS Client has to support the non-LI interface (e.g. Le as shown in figure 4.3.2-1) to the LCS Server (i.e. GMLC). The procedures used by the LCS Server to obtain the location of the target UE are the same as the procedures used to obtain the location of a non-target UE.

Because of the use of non-LI functions and the non-LI interfaces in LALS operations, the LALS comes with a higher risk of detectability and therefore, the LEA and the CSP need to take the following into consideration while employing LALS.

Two categories of detectability that the LI is concerned with are the following:

- Detectability by the target.
- Detectability by unauthorized 3rd parties.

With respect to the LALS, the detectability by the target is dependent on the positioning method used. The level of detectability risk tolerance may vary by jurisdictions and the special circumstances. The TS 33.127 [3] mentions that the involvement of other CSP's networks in fulfilling LALS operations should be avoided. That may restrict the use of

LALS to the non-roaming scenario only. To use LALS in a visited network a method is required to avoid any involvement target's HPLMN.

4.3.3 Overview of the concepts

4.3.3.1 General

As described in clause 4.3.2, there are two modes of LALS invocation. This clause illustrates those two modes. In all the illustrations, the xIRI/IRI used to report the LALS location reporting is referred to as LALSReport in TS 33.128 [4]. The diagrams do not explicitly show the name of that xIRI/IRI.

4.3.3.2 LALS target positioning mode

In the illustration shown in figure 4.3.3.2-1, LEA issues a warrant on the UE for LALS with immediate or periodic location reporting.

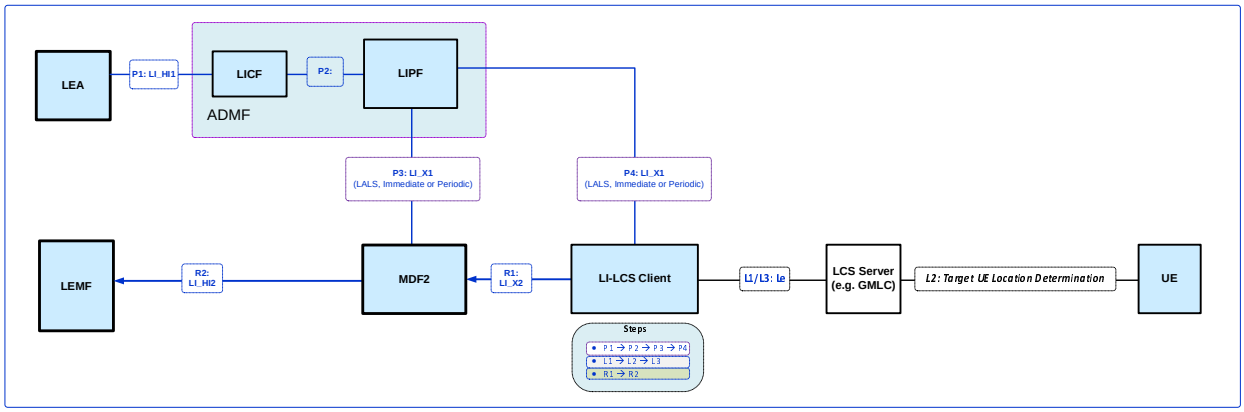


Figure 4.3.3.2-1: LALS target positioning

The LI-LCS Client and the MDF2 are provisioned for LALS (i.e. with the LALS service type). The provisioning of LI-LCS Client includes the additional details needed for the target positioning, e.g. immediate or periodic, with the periodicity if it is the latter. In figure 4.3.3.2-1, P1, P2, P3 and P4 are the provisioning steps.

The LI-LCS Client, based on the parameters received over the LI_X1 from LIPF, interacts with the LCS Server to obtain the target UE's location. The LCS Server interacts with the other network functions (e.g. AMF) to determine the target UE's location using the generic 3GPP LCS procedures and returns the target UE's location to the LI-LCS Client. In figure 4.3.3.2-1, L1, L2 and L3 are the location determination steps. If requested, the LCS Server may provide the intermediate locations of the target UE to the LI-LCS Client.

The LI-LCS Client generates and delivers the xIRI that carries the target's UE location to the MDF2 upon receiving the target UE's location from the LCS Server. The MDF2 forwards it as an IRI message to the LEMF. In figure 4.3.3.2-1, R1 and R2 are the location reporting steps.

For immediate location reporting, the LI-LCS Client requests the LCS Server for target UE's location as and when the LALS is invoked by the LIPF over LI_X1. When periodic location reporting is invoked, the LI-LCS Client requests the LCS Server for target UE's location and then continuously repeats the request at the time interval specified in the LI_X1 task activation. This periodic location reporting continues indefinitely, until deactivated by the LIPF over LI_X1. The cases are illustrated in figure 4.3.3.2-2.

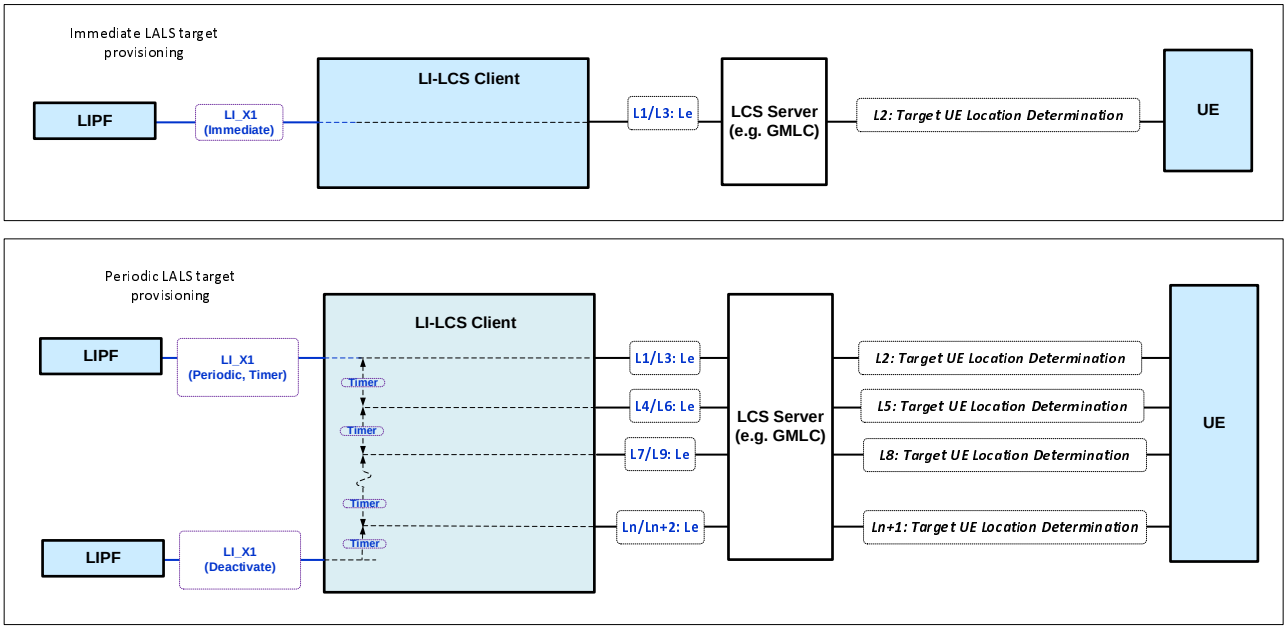


Figure 4.3.3.2-2: Immediate and periodic LALS target positioning

For immediate LALS target positioning, the LI-LCS Client requests the LCS Server for location of the target UE as and when it is provisioned by the LIPF over LI_X1. The steps L1, L2 and L3 are the location determination steps in figure 4.3.3.2-2. The LI-LCS Client self-deactivates the invocation after the location is obtained and reported, or when failure occurs.

For periodic LALS target positioning, the LI-LCS Client starts a timer based on the value received from the LIPF over LI_X1 and at the expiry of the timer, requests LCS Server for location of the target UE and restarts the timer. As shown in figure 4.3.3.2-2, the first request to LCS Server happens when LI-LCS Client is provisioned over the LI_X1 by the LIPF (as in the case immediate LALS target positioning). The LI-LCS Client repeats the request to the LCS Server on each Timer expiry.

In figure 4.3.3.2-2, the L1, L2 and L3 represent the location determination steps for LALS immediate target positioning and [L1, L2, L3], [L4, L5, L6], [L7, L8, L9] and [Ln, Ln+1, Ln+2] represent the location determination steps for LALS periodic target positioning.

4.3.3.3 LALS triggered location mode, LI alongside IRI-POI

In the illustration shown in figure 4.3.3.3-1, LEA issues an interception warrant for the UE with the optional request for LALS triggered location reporting. In this case, the LTF is in the same NF that has an IRI-POI intercepting target events.

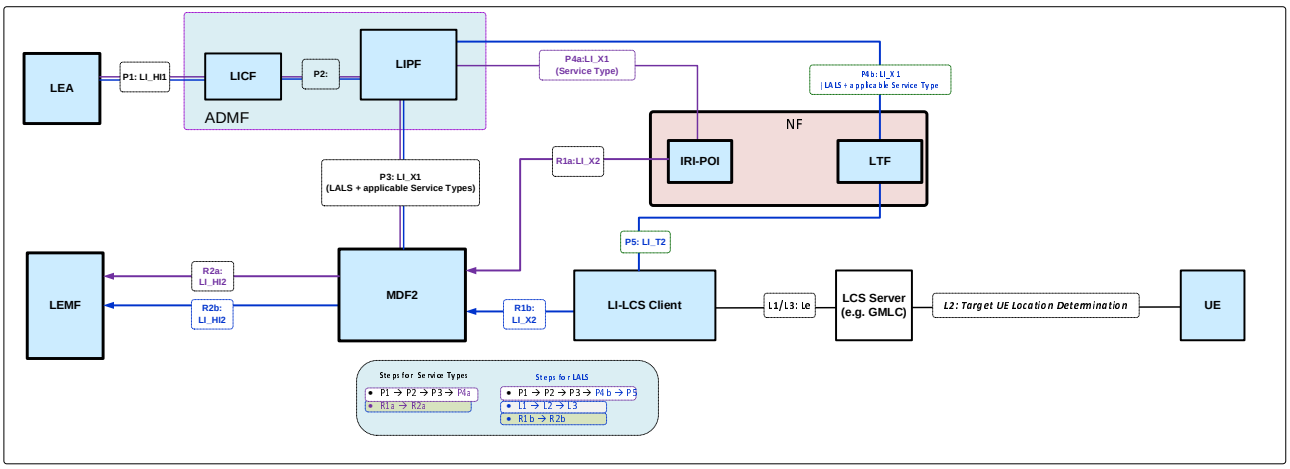


Figure 4.3.3.3-1: LALS triggered location, LTF alongside IRI-POI

The IRI-POI and MDF2 are provisioned for interception of one or more service types (service as applicable to the NF having the IRI-POI) on the target UE. In figure 4.3.3.3-1, P1, P2, P3 and P4a are the provisioning steps for target provisioning for such service type(s) interception.

The LTF in the NF (that has an IRI-POI) and the MDF2 are provisioned with the LALS service type. The provisioning of LTF also includes the service types) to which the LALS applies. Furthermore, the provisioning of LTF includes the additional details that may be needed for the location determination. In figure 4.3.3.3-1, P1, P2, P3 and P4b are the provisioning steps.

The IRI-POI would generate and deliver the xIRIs for the provisioned service type(s) to the MDF2 and the MDF2 would forward those xIRIs as IRI messages to the LEMF. In figure 4.3.3.3-1, the R1a and R2a are those xIRI/IRI reporting steps.

NOTE: The provisioning of MDF2 for LALS and other service type(s) can be combined or an update of the previous one, depending on how the warrant is issued.

When certain events are detected (i.e. particular xIRIs that have the basic location information is generated by the IRI-POI), the LTF (present in the same NF that has the IRI-POI that generated such xIRIs) activates (or invokes) the LALS at the LI-LCS Client over LI_T2, to supplement such xIRIs with the additional location details. The LTF may include a few parameters (received from LIPF over LI_X1) that may be needed for location determination in the LI_T2 trigger. In figure 4.3.3.3-1, P5 represent the step of triggering (viewed as a sort of provisioning).

The LI-LCS Client interacts with the LCS Server to obtain the target UE's location. The LCS Server interacts with the other network functions (e.g. AMF) to determine the target UE's location using the generic 3GPP LCS procedures and returns the target UE's location to the LI-LCS Client. In figure 4.3.3.3-1, L1, L2 and L3 are the location determination steps. If requested, the LCS Server may provide the intermediate locations of the target UE to the LI-LCS Client.

The LI-LCS Client generates and delivers the xIRI that carries the target's UE location to the MDF2 upon receiving the target UE's location from the LCS Server. The MDF2 forwards it as IRI message to the LEMF. In figure 4.3.3.3-1, R1b and R2b are the location reporting steps.

4.3.3.4 LALS triggered location mode, LTF in MDF2

In the illustration shown in figure 4.3.3.4-1, the LEA issues an interception warrant for the UE with the optional request for LALS triggered location reporting. In this case, the LTF is in the MDF2.

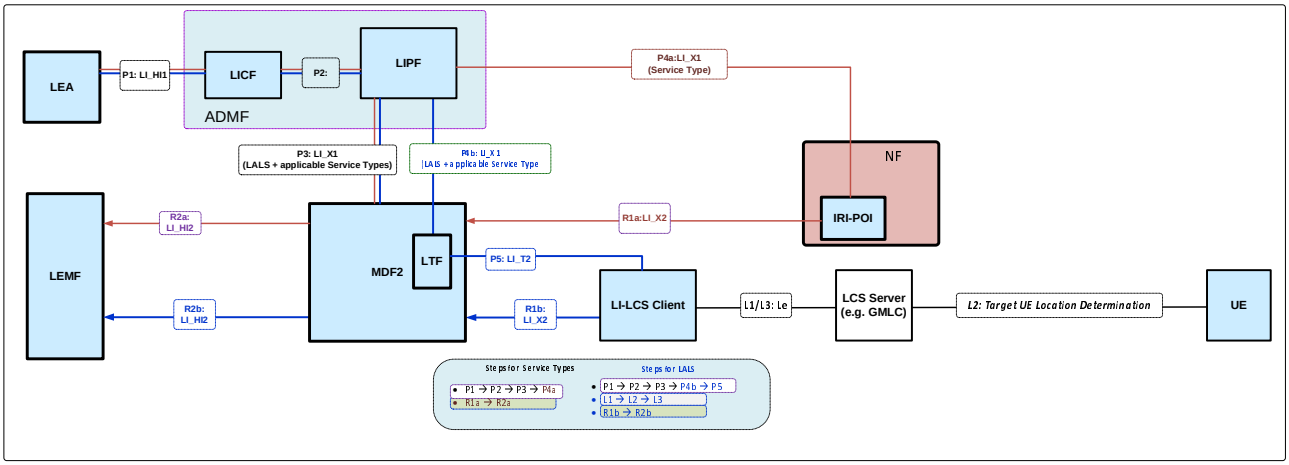


Figure 4.3.3.4-1: LALS triggered location, LTF in MDF2

The IRI-POI and MDF2 are provisioned for interception of one or more service types (service as applicable to the NF having the IRI-POI) on the target UE. In figure 4.3.3.4-1, P1, P2, P3 and P4a are the provisioning steps for target provisioning for such service type(s) interception.

The LTF (in the MDF2) and the MDF2 are provisioned with the LALS service type. The provisioning of LTF also includes the service types) to which the LALS applies. Furthermore, the provisioning of LTF includes the additional details that may be needed for the location determination.. In figure 4.3.3.4-1, P1, P2, P3 and P4b are the provisioning steps.

The IRI-POIs would generate and deliver the xIRIs for the provisioned service type(s) to the MDF2 and the MDF2 would forward those xIRIs as IRI messages to the LEMF. In figure 4.3.3.4-1, the R1a and R2a are those xIRI/IRI reporting steps for the intercepted service types.

NOTE: The provisioning of MDF2 for LALS and other service type(s) can be combined or an update of the previous one, depending on how the warrant is issued.

When certain events are detected (i.e. particular xIRIs are received from the IRI-POIs), the LTF activates (or invokes) the LALS at the LI-LCS Client over LI_T2. The LTF may include a few parameters (received from LIPF over LI_X1) that may be needed for location determination in the LI_T2 trigger. In figure 4.3.3.4-1, P5 represent the step of triggering (viewed as a sort of provisioning).

The LI-LCS Client interacts with the LCS Server to obtain the target UE's location. The LCS Server interacts with the other network functions (e.g. AMF) to determine the target UE's location using generic 3GPP LCS and returns the target UE's location to the LI-LCS Client. In figure 4.3.3.4-1, L1, L2 and L3 are the location determination steps. If requested, the LCS Server may provide the intermediate locations of the target UE to the LI-LCS Client.

The LI-LCS Client generates and delivers the xIRI that carries the target's UE location to the MDF2 upon receiving the target UE's location from the LCS Server. The MDF2 forwards the xIRI as an IRI message to the LEMF. In figure 4.3.3.4-1, R1b and R2b are the location reporting steps.

The implementation of LTF in an MDF2 may be more beneficial when the LALS triggering is required for interceptions of various service types. The approach avoids deploying LTF functionality in every NF that has an IRI-POI.

4.3.4 LALS triggered location scenarios

4.3.4.1 General

The subsequent clauses illustrate a few LALS triggered location scenarios.

4.3.4.2 Scenario 1: LALS triggered location with data interception – LTF in packet core

Figure 4.3.4.2-1 extends the illustration shown in figure 4.3.3.3-1 when the service type to which the LALS applies is Data.

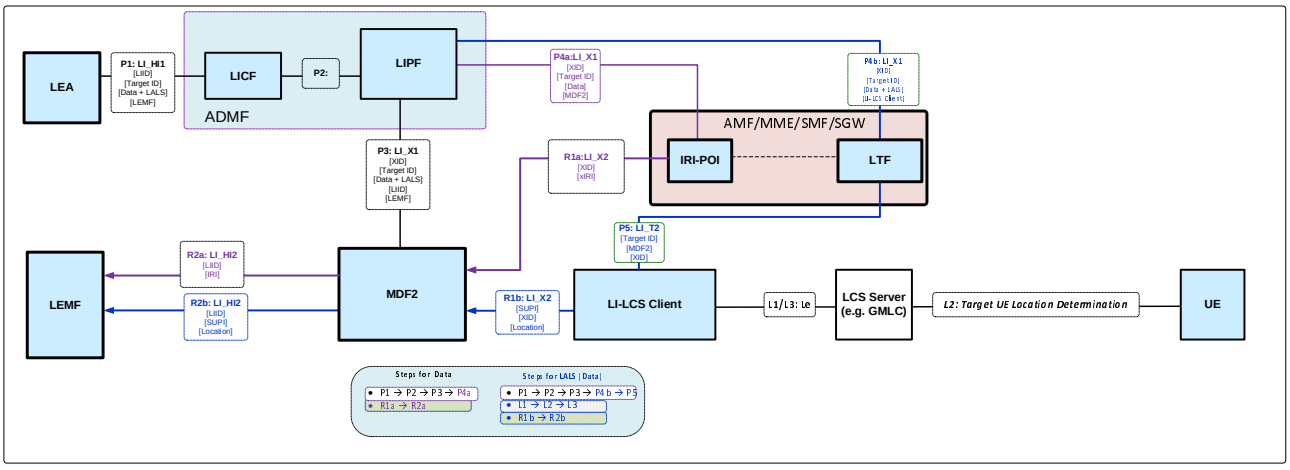


Figure 4.3.4.2-1: Scenario 1: LALS triggered location with data interception – LTF in packet core

The illustration shown in figure 4.3.4.2-1 is a case of a LALS triggered location with the data interception. The LEA issues an interception warrant for Data and LALS. In the illustration, the IRI-POIs located in AMF, SMF, MME, SGW are provisioned for interception with the service type of Data. The LTF present in the same NFs that have those IRI-POIs are provisioned with the service type of Data and LALS. The MDF2 is provisioned with the service type of Data and LALS. The same XID is used in all three of the LI_X1 provisions. In figure 4.3.4.2-1, P1, P2, P3, P4a and P4b are the provisioning steps. As per the provisioning, the LALS triggered location is used for Data interception.

The IRI-POIs generate the xIRIs and deliver the same to the MDF2 over the LI_X2. The MDF2 forwards those xIRIs as IRI messages over LI_HI2. In figure 4.3.4.2-1, R1a and R2a are those xIRI/IRI reporting steps.

The LTF upon detecting particular xIRIs for the packet data interception (with the same XID) would send a LI_T2 trigger to the LI-LCS client. In figure 4.3.4.2-1, P5 represent the step of triggering (viewed as a sort of provisioning). The LI-LCS Client interacts with the LCS Server to determine the target's location. In figure 4.3.4.2-1, L1, L2 and L3 are the location determination steps.

The LI-LCS Client generates and delivers the xIRI that carries the target's UE location to the MDF2 upon receiving the target UE's location from the LCS Server. The MDF2 forwards the xIRI as an IRI message to the LEMF. In figure 4.3.4.2-1, R1b and R2b are the location reporting steps.

4.3.4.3 Scenario 2: LALS triggered location with data + voice interception – LTF in MDF2

Figure 4.3.4.3-1 extends the illustration shown in figure 4.3.3.4-1 when the service type to which the LALS applies is Data and Voice.

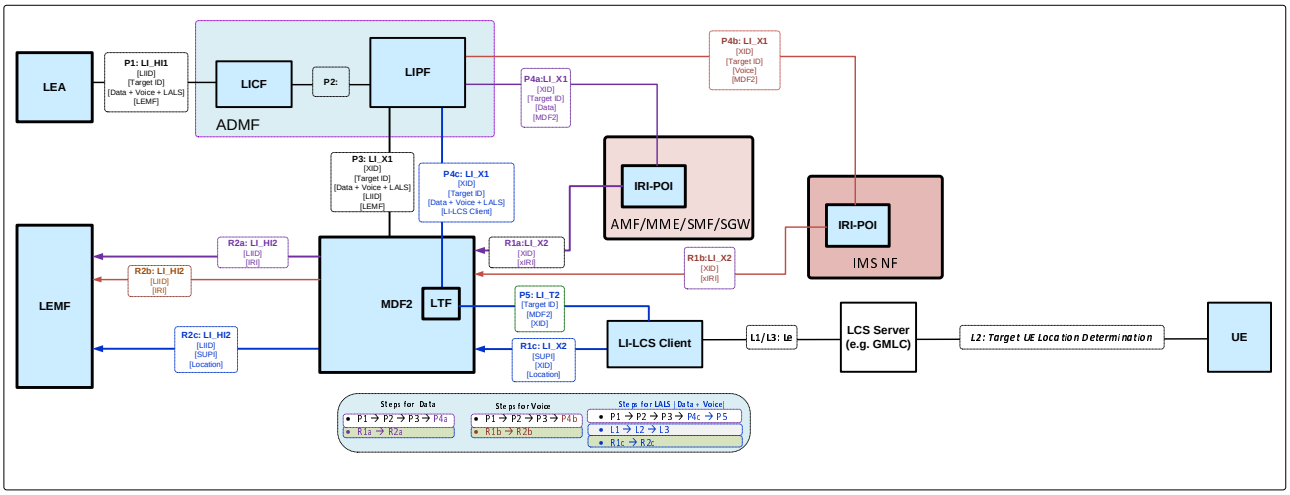


Figure 4.3.4.3-1: Scenario 2: LALS triggered location with data interception – LTF in MDF2

The illustration shown in figure 4.3.4.3-1 is a case of a LALS triggered location with the Data + Voice interception. The LEA issues an interception warrant for Data, Voice and LALS.

In the illustration, the IRI-POIs located in AMF, SMF, MME, SGW are provisioned for interception with the service type of Data. The IRI-POIs in the IMS (e.g. S-CSCF) are provisioned for the interception with the service type of Voice. The MDF2 is provisioned with the service type of Data, Voice and LALS. The LTF in MDF2 is provisioned with the service type of Data, Voice and LALS. The same XID is used in all the LI_X1 provisions. In figure 4.3.4.3-1, P1, P2, P3, P4a, P4b and P4c are the provisioning steps. As per the provisioning, the LALS triggered location is used for data interception and voice interception.

The IRI-POIs in packet core generate the xIRIs for packet data interception and deliver the same to the MDF2 over the LI_X2. The MDF2 forwards those xIRIs as IRI messages over LI_HI2. In figure 4.3.4.3-1, R1a and R2a are those xIRI/IRI reporting steps.

The IRI-POIs in IMS NFs generate the xIRIs for voice interception and deliver the same to the MDF2 over the LI_X2. The MDF2 forwards those xIRIs as IRI messages over LI_HI2. In figure 4.3.4.3-1, R1b and R2b are those xIRI/IRI reporting steps.

The LTF in MDF2 upon detecting the receipt of particular xIRIs (in MDF2) that relate to data and voice interception (with the same XID) would send a LI_T2 trigger to the LI-LCS client. In figure 4.3.4.3-1, P5 represent the step of triggering (viewed as a sort of provisioning). The LI-LCS Client interacts with the LCS Server to determine the target's location. In figure 4.3.4.3-1, L1, L2 and L3 are the location determination steps.

The LI-LCS Client generates and delivers the xIRI that carries the target's UE location to the MDF2 upon receiving the target UE's location from the LCS Server. The MDF2 forwards the xIRI as an IRI message to the LEMF. In figure 4.3.4.3-1, R1c and R2c are the location reporting steps.

4.3.4.4 Scenario 3: LALS triggered location, two warrant scenario, one with LALS – LTF in MDF2

Figure 4.3.4.4-1 illustrates a case of two intercepts with differing service types active on the target with only one requiring the LALS.

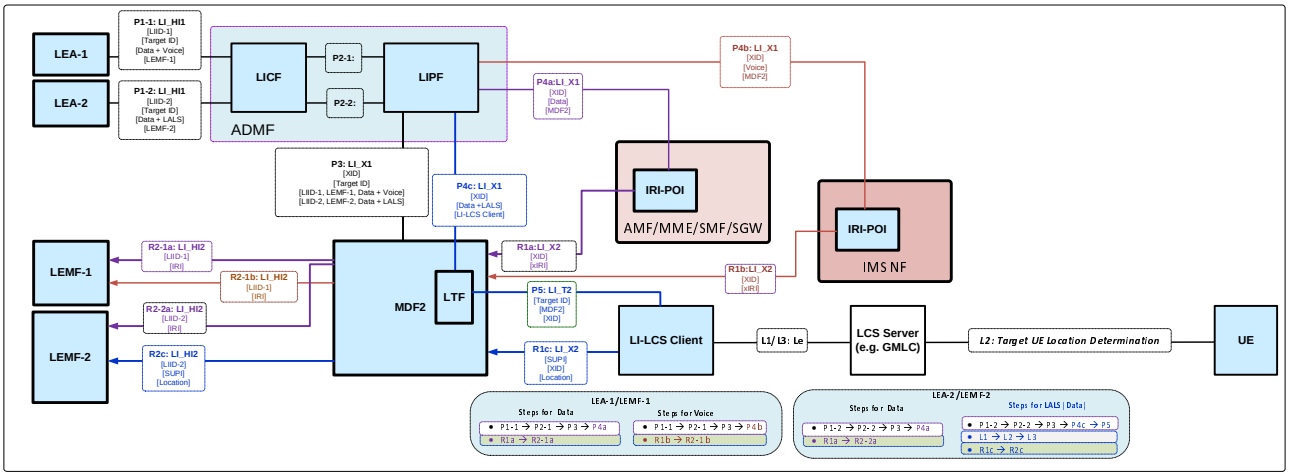


Figure 4.3.4.4-1: Scenario 3 - Two warrants on the target, one requiring LALS – LTF in MDF2

The illustration shown in figure 4.3.4.4-1 is a case where two intercepts are active on the same targe UE. One intercept (from LEA-1) with Data and Voice whereas the second intercept (from LEA-2) is with Data and LALS.

In the illustration, the IRI-POIs located in AMF, SMF, MME, SGW are provisioned for interception with the service type of Data. The IRI-POIs in the IMS (e.g. S-CSCF) are provisioned for the interception with the service type of Voice. The MDF2 is provisioned for the two intercepts with mediation details differentiating the delivery details of the two interceptions. The first one with Data and Voice to LEMF-1 (LIID-1) and the second one with Data and LALS to LEMF-2 (LIID-2). The LTF in MDF2 is provisioned with the service type of Data and LALS. The same XID is used in all the LI_X1 provisions. In figure 4.3.4.4-1, P1-1, P2-1, P3, P4a, P4b are the provisioning steps for the interception warrant from LEA-1 and P2-1, P2-2, add-on P3 and P4c are the provisioning steps for the interception warrant from LEA-2. The step P3 is a modification of the previous step with the mediation details for LEMF-2.

The IRI-POIs in packet core generate the xIRIs for packet data interception and deliver the same to the MDF2 over the LI_X2. The MDF2 forwards those xIRIs as IRI messages over LI_HI2 to LEMF-1 and to LEMF-2. In figure 4.3.4.4-1, R1a and R2-1a and R2-2a are those xIRI/IRI reporting steps.

The IRI-POIs in IMS NF generate the xIRIs for voice interception and deliver the same to the MDF2 over the LI_X2. The MDF2 forwards those xIRIs as IRI messages over LI_HI2 to LEMF-1. In figure 4.3.4.4-1, R1b, R2-1b are those xIRI/IRI reporting steps.

The LTF in MDF2 upon detecting the receipt of particular xIRIs (in MDF2) that relate to data interception (with the same XID) would send a LI_T2 trigger to the LI-LCS client. In figure 4.3.4.4-1, P5 represent the step of triggering (viewed as a sort of provisioning). The LI-LCS Client interacts with the LCS Server to determine the target's location. In figure 4.3.4.4-1, L1, L2 and L3 are the location determination steps.

The LI-LCS Client generates and delivers the xIRI that carries the target's UE location to the MDF2 upon receiving the target UE's location from the LCS Server. The MDF2 forwards the xIRI as an IRI message to the LEMF-2. In figure 4.3.4.4-1, R1c and R2c are the location reporting steps.

4.4 Location acquisition

4.4.1 General

With location acquisition, the authorized LEAs get lawful access to network provided location of target UE on demand. The authorization is done by issuing a warrant for the target UE.

The authorized LEAs send a request to the network for the target UE's location and receive the target UE's location either as a response to the request or as an IRI message, or in both ways. The different options are referred to as location delivery method. The basic principle of location acquisition is illustrated in figure 4.4.1-1 below depicting the three methods of location delivery.

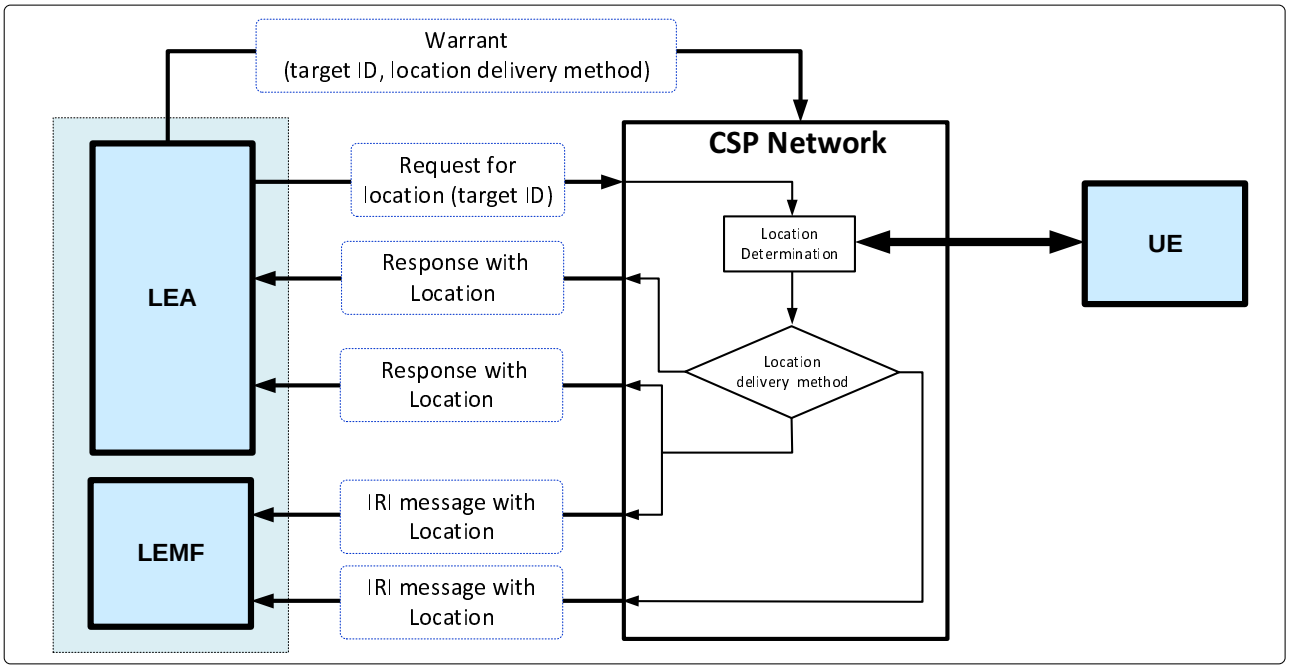


Figure 4.4.1-1: Location acquisition principles

The LEAs provide the Location Delivery Method to the CSP network (along with the warrant) prior to sending the request for the location. The LEAs may request the network for the location as known at the time request is received or request the network for a fresh location.

The details on the scope of reported location, and the protocol (messages/parameters) used are defined in TS 33.127 [3] and TS 33.128 [4]. The location acquisition capabilities do not apply to target non-local ID. The location acquisition capabilities do not apply to the outbound roamers.

Because of the use of non-LI functions in Location Acquisition operations, the Location Acquisition comes with a higher risk of detectability and therefore, the LEA and the CSP need to take the following into consideration while employing Location Acquisition.

Two categories of detectability that the LI is concerned with are the following:

- Detectability by the target.
- Detectability by unauthorized 3rd parties.

With respect to the Location Acquisition, the detectability by the target may exist when the requested Location Acquisition involves some positioning operations. The level of detectability risk tolerance may vary by jurisdictions and the special circumstances. Similarly to LALS, the use of other networks in fulfilling Location Acquisition operations should be avoided. That may restrict the use of Location Acquisition to the non-roaming scenarios only. To use LALS in a visited network a method is required to avoid any involvement of target's HPLMN.

4.4.2 Overview of the architecture

In order to support the location acquisition capabilities, two LI specific logical functions are introduced:

- Location Acquisition Function (LAF).
- Location Acquisition Requesting Function (LARF).

Neither the present document nor the TS 33.127 [3]/33.128 [4] identify/define the NF that can house LAF. The LARF is an LI function within the AMF for 5GS and within the MME for EPS.

The LAF receives the location acquisition request from the LEA. The LARF determines the target UE's location by interacting with the other network functions (i.e. AMF in 5GS and MME in EPS) within the CSP domain. The identities used to identify the target UE are one or more of: SUPI, GPSI, IMSI, MSISDN.

Figure 4.4.2-1 illustrates an overview of the overall architecture for location acquisition methods. All the LI functions and LI interfaces depicted in figure 4.4.2-1 are defined in TS 33.127 [3] and TS 33.128 [4].

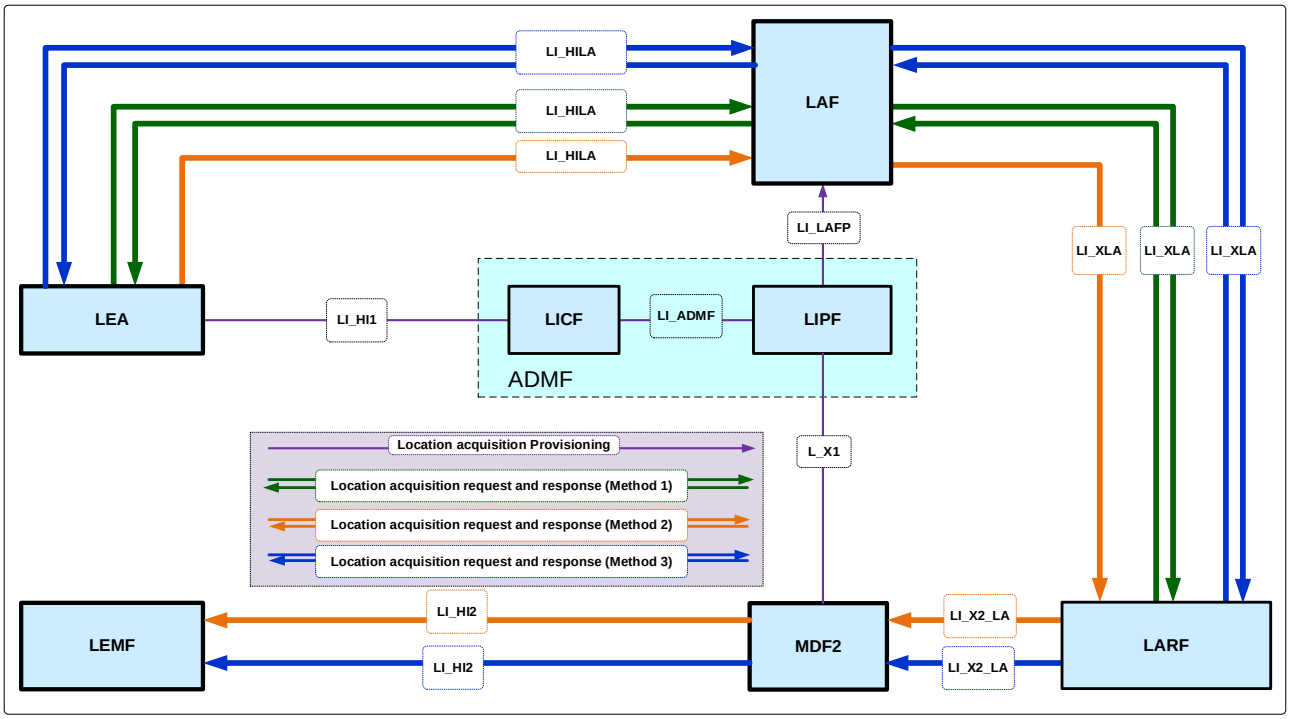


Figure 4.4.2-1: An overview of location acquisition

To realize the location acquisition capabilities, the LEAs have to first issue a warrant that identifies the target UE and location delivery method. As described in clause 4.4.1, two location delivery methods are defined for location acquisition:

- Method 1: as a response to the request.
- Method 2: as an IRI message.
- Method 3: as a response to the request and as an IRI message.

In method 1, the location is returned as a response to the request. In method 2, the location is delivered via the MDF2 to the LEMF as identified in the warrant. In method 3, the location is returned as a response to the request and delivered as an IRI message.

When the warrant is received over the LI_HI1, the LICF/LIPF provisions the LAF over LI_LAFP interface for location acquisition with the location delivery method. For the location delivery method 2 and method 3, the MDF2 is provisioned with the LEMF address for the delivery of the location as an IRI message.

The LEA sends the request for the location acquisition over the LI_HILA interface to the LAF. The LAF upon receiving the request from the LEA forwards the request to LARF over the LI_XLA interface along with the location delivery method. For the method 2 and method 3, the LAF would provide the MDF2 address to the LARF.

The LARF returns the location either to LAF (method 1, method 3) over the LI_XLA interface or to MDF2 (method 2, method3) over the LI_X2_LA interface.

With method 1, the LAF responds to the LEA request with the location over the LI_HILA interface. With method 2, the MDF2 forwards the location to the LEMF as an IRI message over the LI_HI2 interface. With method 3, the LAF responds to the LEA request with location over the LI_HILA interface and MDF2 forwards the location to the LEMF as an IRI message over the LI_HI2 interface.

Within the TS 33.127 [3]/TS 33.128 [4], method 1 is referred to as HILADelivery. method 2 is referred to as HI2Delivery and method 3 is referred to as HILADelivery + HI2Delivery.

4.4.3 Overview of the concepts

4.4.3.1 General

As described in clause 4.4.2, there are three methods of location delivery. This clause illustrates concepts around those three methods.

In the following illustrations, the target identity can be the SUPI or GPSI in the case of 5G and it can be IMSI or MSISDN in the case of 4G. In the response, the SUPI (5G) or the IMSI (4G) and any available other identities of the target UE are included.

4.4.3.2 Location acquisition with HILADelivery method

The illustration shown in figure 4.4.3.2-1, location acquisition concepts when HILADelivery method is used for the location delivery.

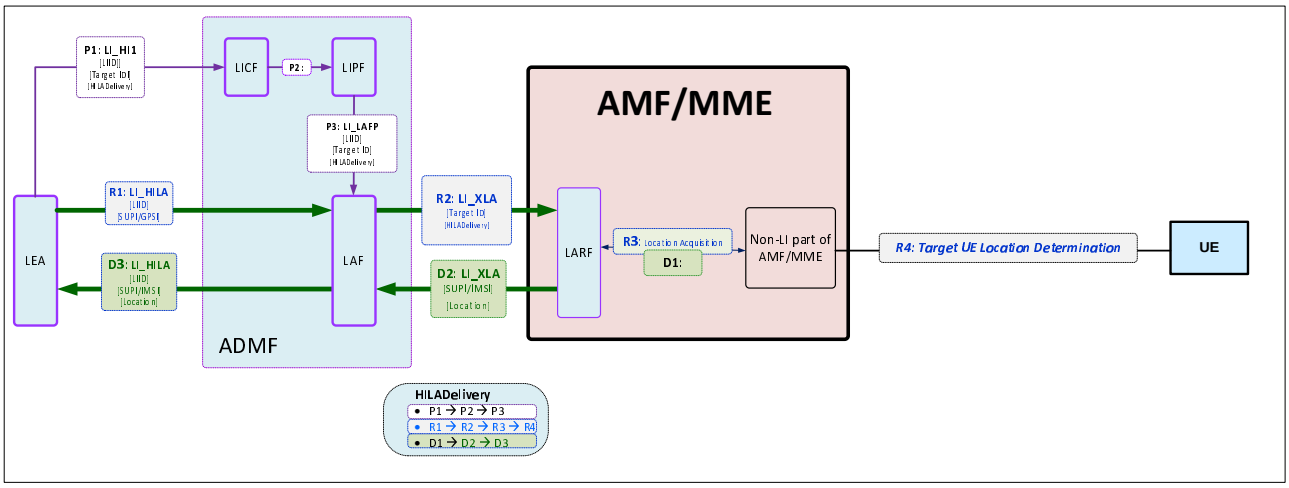


Figure 4.4.3.2-1: Location acquisition with HILADelivery for location delivery method

The LEA issues a warrant on the target UE for location acquisition over the LI_HI1 interface. In the warrant, the LEA indicate the HILADelivery as the location delivery method. The target identity and the LIID are also included in the warrant. The LIPF provisions the LAF over the LI_LAFP interface for location acquisition. The location delivery method (HILADelivery), LIID and the target identity are included in the provisioning. In figure 4.4.3.2-1, P1, P2, and P3 are the provisioning steps.

To get the location of the target UE, the LEA sends the request over the LI_HILA interface to the LAF. The request includes the LIID and the target identity previously provided via the warrant. The LAF identifies the AMF/MME where the target UE (represented by the target identity) is registered.

NOTE: The identification of AMF/MME serving the target UE is not specified in the standard and left for proprietary solutions, which may include the interrogation of UDM/HSS (for non-roaming target UEs), or query of all relevant AMF/MME (for inbound roaming target UEs), or some other means.

After the AMF/MME identification, the LAF sends the request for the location determination to the LARF present in that AMF/MME over the LI_XLA interface. The request includes the target identity and the location delivery method (HILADelivery). The LARF interacts with the non-LI part of AMF/MME for the target location. Depending on the request, the AMF/MME may interact with the other network functions within the CSP domain (e.g. LMF/E-SMLC) to determine the target UE's location. Once the location is obtained, the non-LI part of AMF/MME would return the location of the target UE to the LARF. In figure 4.4.3.2-1, R1, R2, R3 and R4 are the location request steps.

The LARF returns the location over the LI_XLA interface to the LAF (since the location delivery method is HILADelivery). The LAF returns the location to the requesting LEA over the LI_HILA interface. In figure 4.4.3.2-1, D1, D2 and D3 represent the location delivery steps.

4.4.3.3 Location acquisition with HI2Delivery method

The illustration shown in figure 4.4.3.3-1, location acquisition concepts when H2Delivery method is used for the location delivery.

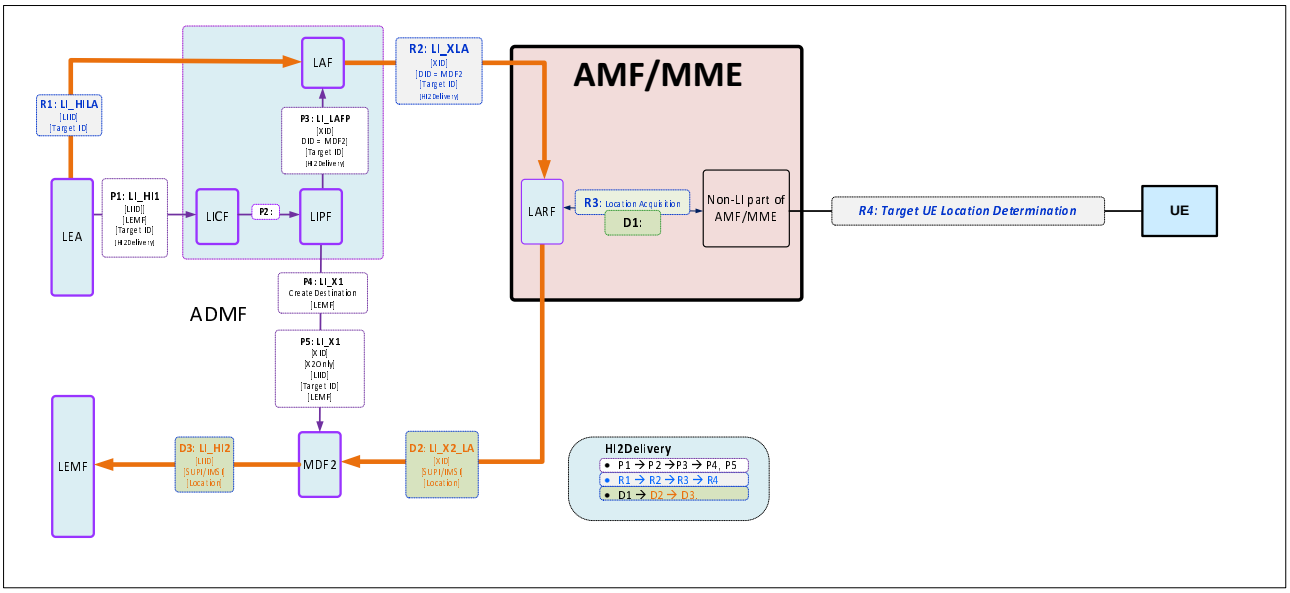


Figure 4.4.3.3-1: Location acquisition with HI2Delivery for location delivery method

The LEA issues a warrant on the target UE for location acquisition over the LI_HI1 interface. In the warrant, the LEA indicate the HI2Delivery as the location delivery method. The target identity, the LIID and the LEMF address are also included in the warrant. The LIPF provisions the LAF over the LI_LAFP interface for location acquisition. The location delivery method (HI2Delivery), MDF2 address and the target identity are included in the provisioning. The LIPF also provisions the MDF2 for location acquisition. The LIID, LEMF address and the target identity are included in the provisioning. Prior to the provisioning, the LIPF creates the LEMF as the destination address for the task at the MDF2. In figure 4.4.3.3-1, P1, P2, P3, P4 and P5 are the provisioning steps.

For location acquisition, the LEA sends the request over the LI_HILA interface to the LAF. The request includes the LIID and the target identifier. The LAF identifies the AMF/MME where the target UE is registered (see the NOTE in clause 4.4.3.2) and sends the request for the location determination to the LARF present in that AMF/MME over the LI_XLA interface. The request includes target identity, location delivery method (HI2Delivery) and the MDF2 address. The LARF interacts with the non-LI part of AMF/MME for the target location. Depending on the request, the AMF/MME may interact with the other network functions within the CSP domain (e.g. LMF/E-SMLC) to determine the target UE's location. Once the location is obtained, the non-LI part of AMF/MME would return the location of the target UE to the LARF. In figure 4.4.3.3-1, R1, R2, R3 and R4 are the location request steps.

The LARF forwards the location over the LI_X2_LA interface to the MDF2 (since the location delivery method is HI2Delivery). The MDF2 forwards location to the LEMF over the LI_HI2 interface. In figure 4.4.3.3-1, D1, D2 and D3 represent the location delivery steps.

4.4.3.4 Location acquisition with HILADelivery and HI2Delivery methods

The illustration shown in figure 4.4.3.4-1, location acquisition concepts when HILADelivery + H2Delivery method is used for the location delivery.

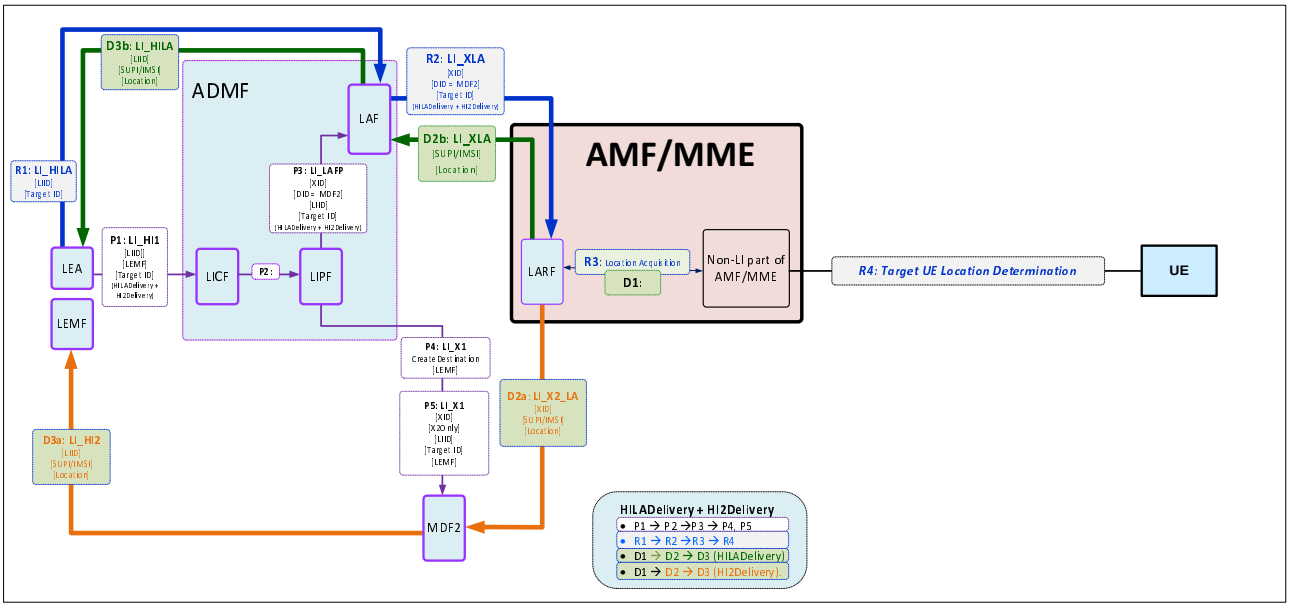


Figure 4.4.3.4-1: Location acquisition with HILADelivery + HI2Delivery for location delivery method

The LEA issues a warrant on the target UE for location acquisition over the LI_HI1 interface. In the warrant, the LEA indicate the HILADelivery + HI2Delivery as the location delivery method. The target identity, the LIID and the LEMF address are also included in the warrant. The LIPF provisions the LAF over the LI_LAFP interface for location acquisition. The location delivery method (HILADelivery + HI2Delivery), MDF2 address and the target identity are included in the provisioning. The LIPF also provisions the MDF2 for location acquisition. The LIID, LEMF address and the target identity are included in the provisioning. Prior to the provisioning, the LIPF creates the LEMF as the destination address for the task at the MDF2. In figure 4.4.3.4-1, P1, P2, P3, P4 and P5 are the provisioning steps.

For location acquisition, the LEA sends the request over the LI_HILA interface to the LAF. The request includes the LIID and the target identifier. The LAF identifies the AMF/MME where the target UE is registered (see the NOTE in clause 4.4.3.2) and sends the request for the location determination to the LARF present in that AMF/MME over the LI_XLA interface. The request also includes the target identity, location delivery method (HILADelivery + HI2Delivery) and the MDF2 address. The LARF interacts with the non-LI part of AMF/MME for the target location. Depending on the request, the AMF/MME may interact with the other network functions within the CSP domain (e.g. LMF/E-SMLC) to determine the target UE's location. Once the location is obtained, the non-LI part of AMF/MME would return the location of the target UE to the LARF. In figure 4.4.3.4-1, R1, R2, R3 and R4 are the location request steps.

Since the HI2Delivery is one of the two location delivery methods, the LARF forwards the location over the LI_X2_LA interface to the MDF2. The MDF2 forwards location to the LEMF over the LI_HI2 interface. In figure 4.4.3.4-1, D1, D2a and D3a represent the location delivery steps.

Since the HILADelivery is one of the two location delivery methods, the LARF also returns the location over the LI_XLA interface to the LAF. The LAF returns the location to the requesting LEA over the LI_HILA interface. In figure 4.4.3.4-1, D1, D2b and D3b represent the location delivery steps.

Annex A:
Change history

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2024-07	SA3#94-LI	s3i240510				Initial version	0.0.1
2024-11	SA3#95-LI	S3i240754				Revised as agreed by SA3LI	0.0.2
2026-07	SA3#102-LI	S3i260362				Agreed for Change Control	0.0.3
2026-08	SA#113	SP-260733				Submitted to SA#113 Plenary for Approval	1.0.0
2026-09	SA#113	-				Approved by SA#113 Plenary for Change Control	19.0.0

History

Version	Date	Status
V19.0.0	October 2026	Publication